

Intra oral radiographic projections & techniques

A radiographic image is a type of shadow. The radiation source for this shadow is the focal spot, and the film records the shadow.

1. Periapical Radiography

The name "periapical" originates from the Greek words *peri* (meaning "around") and *apical* (meaning "tip"). This view is utilized for both anterior and posterior teeth.

Objective:

The goal is to capture the tip of the root (apex of the tooth) on the film. This helps dentists visualize the tooth and the surrounding bone, which is often useful in finding the cause of pain in a specific tooth.

Uses and Indications:

- **Indications:**
 - Detection of apical infection.
 - Assessment of periodontal status after trauma to the teeth and alveolar bone.
 - Assessment of the presence and position of un erupted teeth.
 - Assessment of root morphology.
 - During endodontic procedures.
 - Presurgical implant insertion bone evaluation.
- **Other Uses:**
 - To determine the need for endodontic therapy.
 - To visualize the successful progression of endodontic therapy after it has been initiated.
 - For detection of supernumerary teeth & impacted teeth.

Techniques:

There are two commonly used intraoral techniques:

- **a/ Bisecting technique:** This is the older and the easier of the two techniques.
 - **b/ Parallel technique:** Originally developed by MC Cormack, the results of this technique are superior to those of the bisecting one.
-

Theory of Parallel Technique

It is named this because the film and the tooth must be parallel to each other.

Requirements:

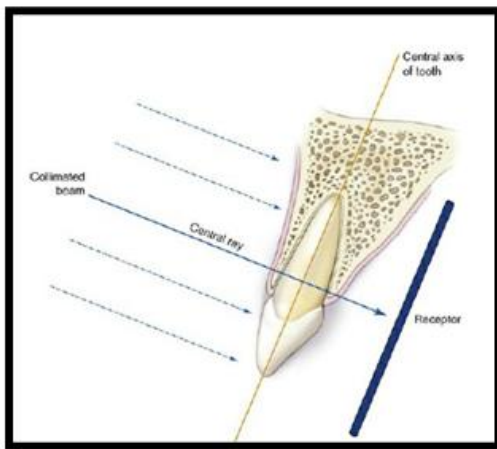
1. The target-object distance must be as long as possible and practical.
2. The X-ray must strike the object (tooth) and the film at a right angle.
3. The film must be placed parallel with the plane that passes through the long axis of all teeth being examined.

This last requirement means there must be a fairly wide separation between the tooth and the film. This separation would cause significant distortion (magnification) if a short target-object distance was used. However, using an extended long cone of 16 inches increases the target-object distance, which compensates for the distortion and unsharpness resulting from the increased object-film distance.

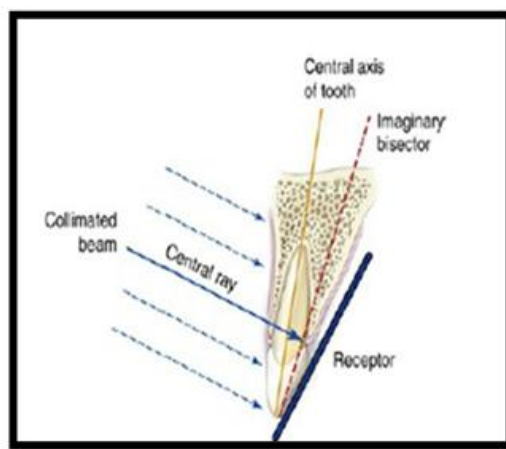
Theory of Bisecting Technique

1. The operator imagines an "imaginary bisector" of the angle formed by the long axis of the tooth and the long axis of the film. This angle is created where the film touches the tooth crown.
2. The operator directs the central ray of the beam through the tooth's apex so the central ray strikes the bisector at 90°. If done properly, this angulation results in a tooth image that is the exact length of the object.

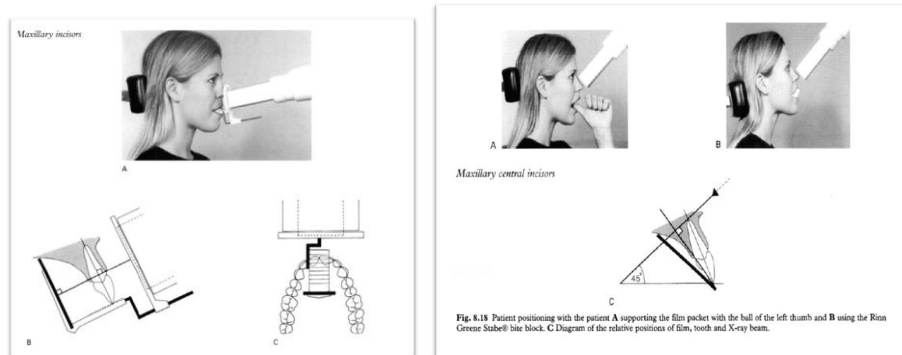
In this technique, the film is in contact with the tooth crown, meaning the tooth and film are not parallel. As a result, all areas above and below the apex are distorted. This distortion can be reduced by using a long cylinder, as a longer distance between the radiation source and the object makes the rays more parallel.



Parallel technique



Bisecting technique



General Concepts for Periapical Radiography

Identification Dot:

- This is a round, raised dot found in the corner of each film.
- It permits rapid and proper film orientation and placement.
- The manufacturer orients the film so the convex side of the dot faces the front of the packet and the radiation source.
- During exposure, the dot should be placed 2-3 mm away from the incisal or occlusal surface.

Horizontal and Vertical Angulations:

- 1. Horizontal angulation: Refers to the X-ray beam's direction in a horizontal plane.
- 2. Vertical angulation: Refers to the X-ray beam's direction in a vertical plane.
 - **Plus vertical angulation:** The beam is tipped downward.
 - **Minus vertical angulation:** The beam is tipped upward.

Angulations Guidelines for Bisecting Angle Projections: (Assumes the occlusal plane is oriented parallel with the floor)

Projection	Maxilla	mandible
Incisors	+ 40 degree	- 15 degree
Canines	+ 45 degree	- 20 degree
Premolars	+ 30 degree	- 10 degree
Molars	+ 20 degree	- 5 degree

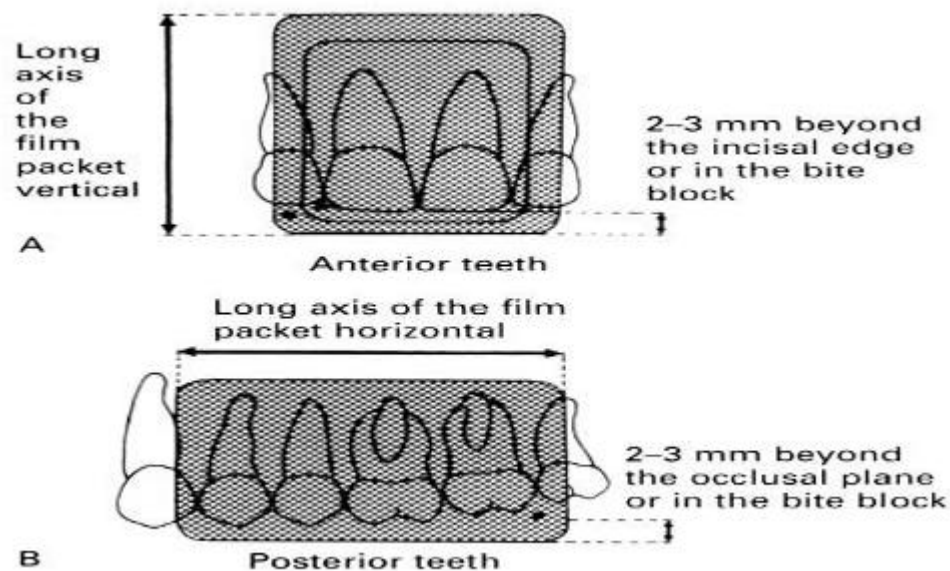
Quality Standards for Periapical Radiography:

- The image must be well defined, with no blurring.

- **Correct anatomical coverage:** The film must show all the tooth/teeth of interest, including the crown and root(s).
- The proximal surfaces of the teeth should not overlap.
- There should be no "coning off" (no blank areas on the radiograph).
- The anatomical area, the apex of the tooth, and 2-3mm of surrounding bone must be shown to allow for an assessment of the apical anatomy.

Periapical Film Placement:

- **Anterior teeth:** The long axis of the film packet is vertical. It should be 2-3 mm beyond the incisal edge or in the bite block.
- **Posterior teeth:** The long axis of the film packet is horizontal. It should be 2-3 mm beyond the occlusal plane or in the bite block.



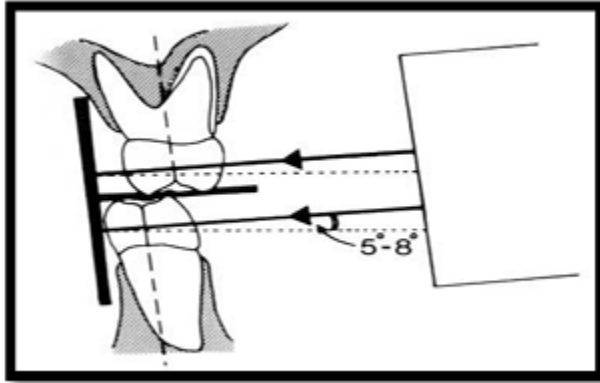
2. Bitewing Radiography

Diagnostic Objectives of Bitewing Radiography:

- Detect early interproximal caries before it becomes clinically apparent.
- Detect secondary caries below restorations.
- Assess loss of the interdental and furcation bone.

Film placement and angulations for bitewing films:

- Bitewing X-ray film is used to reveal interproximal caries and visualize periodontal-condition.
- **Adults:** Need 2 bitewing films on each side of the jaw (premolar and molar areas).
- **Children (12 years old):** Need one film on each side.



- **Technique:**
 - The patient is positioned with the occlusal plane horizontal.
 - The film's tab is placed on the occlusal surfaces of the lower teeth.
 - The patient is asked to close their teeth firmly together on the tab.
 - The beam is aimed directly through the contact areas, at right angles to the teeth and film in the horizontal plane.
 - The beam is aimed at an approximate 5°-8° downward vertical plane, which compensates for the curve of Monson.

3. Occlusal Radiography

Diagnostic Objectives of Occlusal Radiography:

- To locate supernumerary, un erupted, and impacted teeth.
- To localize foreign bodies in the jaws and floor of the mouth.
- To identify and determine the full extent of disease (e.g., cysts, osteomyelitis, malignancies) in the jaws, palate, and floor of the mouth.
- To evaluate and monitor changes in the midpalatal suture during orthodontic palatal expansion.
- To detect and locate sialoliths (salivary stones) in the ducts of sublingual and submandibular glands.

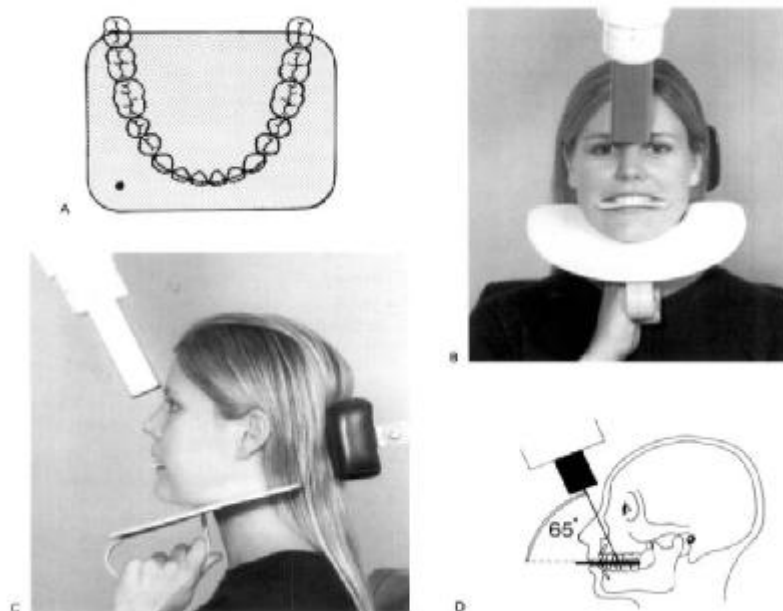
Types of occlusal projection:

1. Maxillary occlusal projections
2. Mandibular occlusal projections

Maxillary Occlusal Projections

a- Upper standard occlusal

- **View:** Shows the anterior part of the maxilla and the upper anterior teeth.

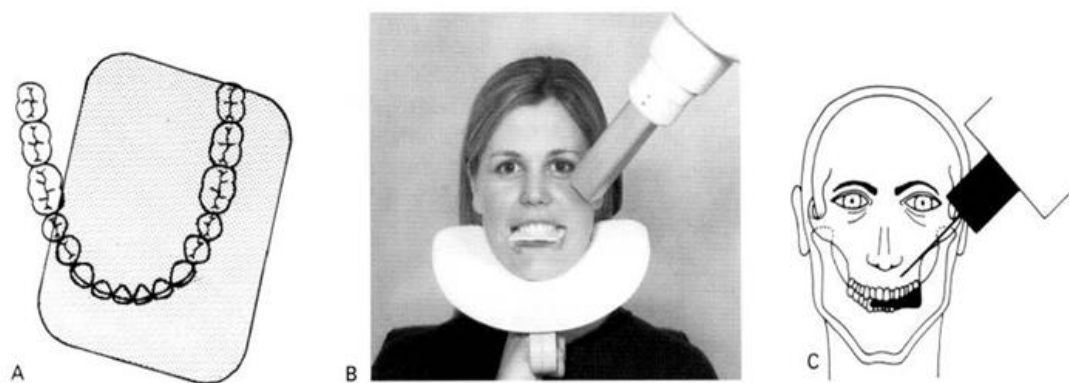


- **Technique:**

1. Patient's occlusal plane is horizontal and parallel to the floor.
2. Film is placed on the occlusal surfaces of the lower teeth, centered in the mouth with the long axis crossways. The patient bites together gently.
3. X-ray tube is positioned above the patient in the midline, directed downward through the bridge of the nose at 65°-70° to the film packet.

b- Upper oblique occlusal

- **View:** Shows the posterior part of the maxilla and the upper posterior teeth.



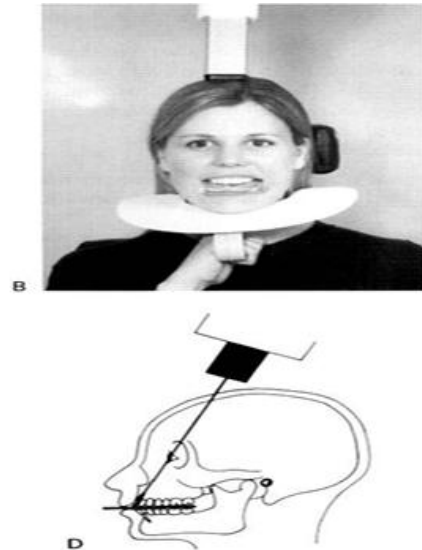
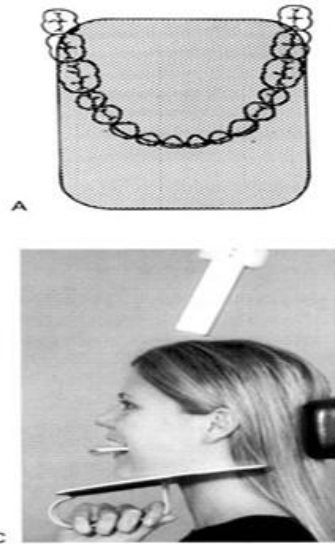
- **Technique:**

1. Patient's occlusal plane is horizontal and parallel to the floor.
2. Film is placed on the occlusal surfaces of the lower teeth, positioned to the side of the mouth being examined, with the long axis anterior-posterior. The patient bites gently.

3. X-ray tube is positioned at the side of the patient's face, directed downwards through the cheek at $65-70^\circ$ to the film.

c- Vertex occlusal

- **View:** Shows a plan view of the maxilla's teeth-bearing area from above.
- **Use:** To assess the bucco-palatal position of un erupted canines.



- **Technique:**
 1. Patient is seated with the occlusal plane horizontal and parallel to the floor.
 2. Film is placed on the occlusal surfaces of the lower teeth with its long axis anteroposteriorly. The patient bites on to it.
 3. X-ray tube is positioned above the patient in the midline, directed downwards through the vertex of the skull.

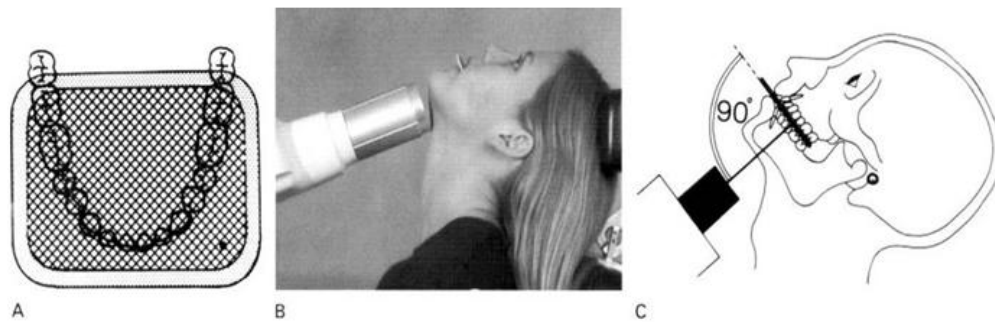
Mandibular Occlusal Projection

Types:

- a/ Lower 90° occlusal (true occlusal)
- b/ Lower standard occlusal
- c/ Lower oblique occlusal

a/ Lower 90° occlusal (true occlusal)

- **View:** Shows a plan view of the mandible's tooth-bearing area and the floor of the mouth.

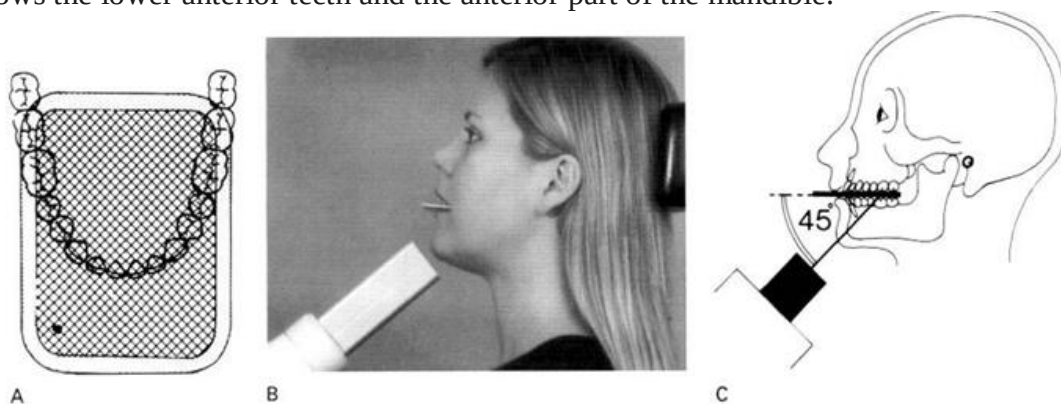


- **Technique:**

1. Patient tips their head backward as far as is comfortable and is supported.
2. Film is placed centrally in the mouth on the occlusal surfaces of the lower teeth, with the long axis crossways. The patient bites gently.
3. X-ray tube is placed below the patient's chin in the midline, centering on an imaginary line joining the first molar at 90° to the film.

b/ Lower standard occlusal

- **View:** Shows the lower anterior teeth and the anterior part of the mandible.

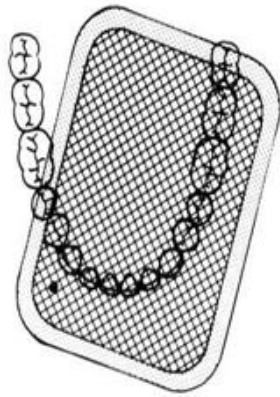


- **Technique:**

1. Patient is seated with the head supported and the occlusal plane horizontal and parallel to the floor.
2. Film is placed centrally in the mouth with the long axis anteroposterior. The patient bites on the film gently.
3. X-ray tube is positioned in the midline, centering through the chin point at 45° to the film.

c/ Lower oblique occlusal

- **View:** Shows the submandibular salivary gland on the side of interest.



A



B



C

- **Technique:**

1. Patient's head is supported, raised, and rotated away from the side under investigation.
2. Film is placed on the occlusal surfaces of the lower teeth, over to the side being investigated, with the long axis anterior-posteriorly. The patient bites on the film gently.
3. X-ray tube is directed upwards and forwards toward the film from below and behind the angle of the mandible, parallel to the lingual surface of the mandible.